ActInf GuestStream 082.4 ~ Robert Worden "Assessment of the Brain Wav

GuestStream | Robert Worden | 1:37:56

hello and welcome everyone this is
active inference guest stream number
82.4 on July 29th
20124 We are continuing this series of
presentations and discussions with
Robert Warden and friends so today we'll
be discussing assessment of the brain
wave hypothesis thank you Rober and
everyone else for joining so please for
the
presentation okay um this talk is about
a hypothesis that we've had previous
live streams about the hypothesis is
that there is a wave in the brain and
this talk is assessing the likelihood of
that being true basically and so if you
want to see the previous talks which one
of which was about the link between the
wave and Consciousness then you can
follow this link here to an active
inference web page so what I'm going to
in this talk is firstly try and describe
to you what the wave hypothesis is and
secondly in point two give some of the
evidence which I think points towards
there being a wave in the brain and some
of the evidence against it uh now the
question then arises we've got different
lines of evidence how do we combine
those lines of evidence to make some
overall estimate of How likely it is
that there is a wave in the brain and so
what I do in point three stage three is
at an overall basian assessment of the
probability that there is this wave and

the answer I come to is about 0.5 is the
probability the detailed number doesn't
matter what matters is that it's not a
tiny probability in other words we can't
just dismiss this wave hypothesis out of
hand we need to think seriously about
what it might mean for the brain If
there is a wave and at that point for um
we will break for discussion and and
feedback from people on the court
and after point 4 we'll go on to discuss
possible future tests of the wave
hypothesis in five and finally after
five further discussion again so that's
the plan of the talk roughly um and the
first part is to describe what the wave
hypothesis is so wave hypothesis says
that there is some region of the brain
which is approximately round in which
there are waves going in all three
Direction c s and there's a round region
shown with a couple of waves and these
waves represent objects in space so the
long wavelength wave Lambda 2 the green
one represents an object at R2 a small
distance and the Red Wave with shorter
wavelength represents an object at
longer distance so that is the
hypothesis that there is this wave
region in the brain and neurons couple
to the wave as transmitters and
receivers so a wave acts as a kind of
working memory for a 3D model of space
and all the models around the animal in
space at the time so that in essence is
the wave hypothesis but the important
thing about it is how well would a wave
based spatial memory

perform and you do some Elementary Mass
the number of waves you can get in this
round region it's approximately if
there's some minimum wave length which I
call Lambda in then the number of waves
you can get in the in the space is D
over Lambda Cub roughly so that's quite
a lot of things you can represent in the
wave and the spatial Precision you can
get is of the order of Lambda Min over D
and that can be quite small so the
benefits of this wave storage are you
can get high precision and that I claim
you can't get out of neurons you can get
high capacity to represent lots of
different objects you get low spatial
distortion
and generally it is a very simple and
direct physical representation of 3D
space it really acts like a hologram so
the question is is there any evidence
that there is such a wave in the brain
and I think there are three kinds of
three main pieces of
evidence the first one comes from the
mum Thalamus and the second one comes
from the central body of the insect
brain now the pamus is shown in the left
in green for three
mammals the point about the thus is it's
approximately round so it's well suited
to hold a wave and it has all the right
connections to hold a representation of
space it has all the sense data coming
in pretty much all the sense data and it
has connections to all the cortical
regions that are making sense of that
sense

data so there are more specific reasons
about the phalus but I'll come to those
later so that's why the thalamus is a
likely can holding the wave and
similarly in insects there's a central
body of the insect brain which is a
small part of the center of the insect's
brain which its shape is remarkably
preserved across all insect species and
it's nearly round so the central body
like the phus is well suited to hold the
wave and it's connected has all the
right neural connections to holder
representation of space so those are the
first two pieces of evidence now the
third line of evidence comes from
building neural computational models and
it's a thing that I think has not been
much remarked
upon um 3D spatial com cognition is very
fundamental it's the first thing an
animal has to do to understand the space
around it in order just to
move now I've tried building 3D models
of space at M David marah's level too
and the point is that to build a model
of space in the brain it has to be
fairly precise a approximately 1% and
you can see this because vision and
other sense data is fairly precise and
there be no point in getting that
precise sense data if you didn't make a
a fairly precise model of space out of
it however it's very difficult to do
that with neurons because in a fifth of
the second which is approximately the
time an animal has to have to construct
his model of space a typical neuron

Spike train has only about 10 spikes in
it and so this can represent a real
number such as a dimension in space with
a Precision of approximately square root
one in square root 10 parts in other
words 1 and three or 30% and that
Precision is very far out from the
Precision which I think you need to
represent 3D space I think why this has
not been recognized or not been spotted
is that we've done neural modeling of
the brain for years and years but
everybody models neurons as highly
precise
reproducible uh units with outputs of
real variables as is shown in the top of
this figure the output of a neural
neuron is usually model as a real
variable with very high Precision so you
can get high spatial Precision out of
that whereas real neurons they have
Spike trains and in a short time a spike
train doesn't represent a number
precisely at all so this I think has not
been spotted
uh and I think it makes a huge gap in
neural models of the
brain and as I've just said the wave
storage can get much better Precision so
wave storage might fill this Gap and
that is the third piece of evidence for
a wave in the brain so in summary three
pieces of three main pieces of evidence
beamus the insect central body and these
neural computational models which
haven't model model 3D space properly
and I think that's a bit of an
embarrassment for the neural model of

the brain really so uh if you have those
pieces of evidence and some evidence
against the wave which I'll come to uh
how do you assess the overall
probability that the wave is there well
it's a binary hypothesis either there is
a wave or there's not a wave and we want
to estimate the probability that there
is one based on several pieces of
evidence and this in analogy is like a
biased coin in other words How likely is
that coin going to turn up heads on the
next time we toss it and you do various
experiments one experiment is toss the
coin M times and you get heads end
times so what you do is several
different experiments you might toss
with your left hand might toss the coin
from an airplane and you might build a
special coin tossing machine and with
each one you do some experiments and so
the best estimate coming out of all
those experiments is the sum of all the
heads over the sum of all the coin
tosses
and so this is the way you the basian
way that you combine the evidence from
different lines of evidence on the same
hypothesis you a lot a weight to each
one which has a certain subjective
element to it and then you sum them all
according to their weights and that's
what I'm going to do to make a basin
combination of the different lines of
evidence about the wave hypothesis and I
just what you do is you
simply you a lot coin tosses to each
line of evidence and you SU the coin

toses
together that I think is uh how people
do science roughly and it is if you like
it's in the textbooks as the basian
philosophy of
science so when we do this assessment
what answer do we get and I've
summarized the answers in this little
table and I've put red rows for things
that are mainly against the wave
hypothesis and green rows for things
that are mainly for it and the two red
rows you'll find are prior
considerations our prior prejudices and
the idea and biophysical mechanisms and
I've given those large negative
weights I've done that deliberately to
show that the overall result is not too
sensitive to those negative weights you
can adjust them a bit and you won't
change the result much and the positive
lines of evidence are the three things I
talked about the thalamus the insect
central body and computational models
and a couple of others there is some
evidence from phenomenal Consciousness
and there are some other
considerations so what I'm going to do
quite briefly is go through those
different rows in the table summarizing
basically what the considerations are I
won't spend too long on them because the
tables are not that interesting but you
can come back to anyone or you can stop
me on anyone ask questions if you like
or come back to anyone in discussion so
so what is the prior considerations that
go against the wave hypothesis first one

I've put in a big weight of 20 for oam's
razor in other words it's an extra
hypothesis we just don't want an extra
hypothesis so let's put a weight against
it and so I've put a big weight of 20
against that it's bigger than I think is
realistic but I've done so purely to
make it clear that the eventual answer
is quite robust against that
and there's a couple of other arguments
I've put in which people might make one
is the classical neural model of the
brain has served as well so why do we
need anything else but I think that's a
weak line of argument because the
classical neural model of the brain
hasn't worked that well that we're ready
to that we've described all the brain
there's still massive number of things
we don't know about the brain so that's
a weak argument and the second argument
is well we never seen a wave so surely
that means they're not one but that that
again as a weak argument I think because
we just don't know how to look for this
wave and it probably has very low
intensitive reasons I'll come to later
so those are weak arguments but
nevertheless I come up with 25 points
against the wave 25 Point against the
wave from prior
considerations so now we move on to
positive in evidence for the wave and
the first one of these is the mamalian
thalamus there is both General evidence
which I've talked about in other words
the phalus has the right shape and it's
in the right connect has the right

connections to all the three 3D model
ens and there's what I call Smoking Gun
gun evidence something that's much more
specific about the thalamus which points
to their being a waven now I'm going to
come to that later in the talk so I
won't go into that detail in detail now
but in summary I think Thalamus offers
quite a lot of positive support 14
against two for the
wave then we come on to the insect
central body there is again the general
evidence that the insect central body
has the right shape and it's highly
conserved across all insect species now
I think that is quite significant in
other words every insect in the world
has the same shape pretty much the same
shape of its central body so the shape
must mean something whereas you compare
that with other parts of the insect
brain like the mushroom bodies for
instance they have very variable
shapes and they just have to make neural
connections but this is a clue that the
central body of the insect brain is
doing something other than making neural
connections and the idea of a wave in it
fits it quite
well um and I think there's other
evidence from the from the insect brain
mainly that the insect brain has very
few neurons in it so there are probably
not any neurons to make a decent
representation of 3D space but we knew
know that insects do it rather well so
that's the central
body next is the computational

modeling as I said 3D spatial cognition
is a vital function of the brain but
that as far as I know Carl can put me
right on this there is no working neural
computational model of it especially
when you ask for a model Which models
Spike trains rather than idealized
theoretical
neurons and in any subsec time intervals
neurons give a big error rate about 30%
whereas we really need 1% or something
to con construct decent spatial
model so the evidence yeah may I ask
just a Precision here the the point here
I think if I follow you well that we
should all bear in mind is the model
that is
needed performance in real time so what
you are saying is the lack of precision
make come also because of some time
pressure that for the to be valid should
be able to build and update this 3D
model of space in quote unquote real
time correct yeah very short time when
you need that model of space to be
really up to date so you only have a in
mammals you have a third of a second and
insects you have even less time and in
that time neurons real neurons are very
imprecise that's the
argument okay so that's the three lines
of evidence for the wave um one line of
evidence I think that goes against is
biophysical mechanisms in other words
how does this wave work and as far as I
know no biophysical model for the wave
has ever been proposed and we make again
the rather pessimistic assumption that

the chances that we'll find one are
small will come to that later but we've
put 10 points against the chances of
finding a decent biophysical model of
the wave now I think that is very
pessimistic especially in view of the
the point that evolution is clever than
we are and so even if we can't think of
it Evolution may be able to think of it
and there in in this table there's some
other biophysics evidence which
approximately balances out so the
biophysics evidence is taken to go
against the wave
broadly uh uh finally the last thing
I'll consider is phenomenal
Consciousness and this was on the last
live stream that if there is a wave in
the brain it leads to a new theory of
how Consciousness works and you can see
that on on that live stream and this
theory is I think is quite nice it
solves a lot of the problems of neural
theories of Consciousness but we don't
want to put too much weight on it
because Consciousness theories are so
controversial but I think what we can do
is that speci
Consciousness the the conscience that we
have right now of the space around us
must come from something happening in
the
brain and our spatial Consciousness is
quite detailed and precise and so that
tells us however it's done that the
model of space in the in the human brain
is quite detailed and
precise and so that again if you come

back to the point neurons can't do it
but way might be able to do it that is
further evidence in favor of the wave
but again because Consciousness is
controversial I haven't given it big
weight at all I've given it four points
so when we and there's some other
evidence that I W don't think I'll go
into so when we combine all the all the
evidence here's the scores we've been
keeping been adding up here's the coin
tosses which came out heads here's the
coin tosses which came out tails and we
have a total of 91 coin tosses of which
47 came out heads so the basically we
were deliberately pessimistic in
stacking up the negative evidence
against the wave but in spite of that
the positive evidence is
one and quite literally the many pieces
of positive evidence add up and so we
get to probability 0.52 or if we take a
simpler view that we just give
every line every piece of evidence one
vote we get 31 votes and 24 of those are
in favor of the way so we get a more
positive number out of that and my main
point is not the number 0.5 or whatever
it is but the the main point is it would
take very strong NE negative evidence to
change this and I don't know of that
evidence and so my main point is really
you can't easily dismiss this wave
hypothesis and if you can't dismiss it
then it raises some big questions
I think if the wave exists we need to
really rethink how the brain works it's
a very different model of the brain we

have um and if not by the way there's
still a huge unsolved problem because
how do these imprecise neurons do 3D
spatial
cognition so
um um what I'm doing the trouble with
the one trouble with the wave hypothesis
is it really rests on the work of one
person at the moment that's just me
that's been interested in it but science
requires reproducibility and debate
between lots of people so what I'm
asking people to do watching this live
stream and asking you to do is please
look at these ideas and assess them for
yourselves uh please provide some
commentary of your own on the wave
whether it's Specialized or generalized
or whatever please challenge the ideas
please get a hold of the spreadsheet you
can download the spreadsheet and and
change the way Waits and see what you
end up with or please bring new evidence
to Bear there may be some evidence
against the wave for it that I don't
know about and please agree or disagree
and what I'd like to do is to collect
that commentary and to make that
commentary available when people want uh
on this website you can get the
spreadsheet and please therefore send
your comments to my
email so that's the call for commentary
and we come on now oh by the way this is
very important Carl friston has endorsed
the call for commentary I hope you're on
the call Carl um based on this short
description of the wave which I'll leave

you to read Carl writes this is the
basian waves in a brain hypothesis and
it challenges conventional views of
belief updating in the basian brain and
it offers a perspective on neuromorphic
computation that's supported not only by
theoretical accounts of functional brain
architectur but receiving empirical
evidence that speaks to a fatic
mechanisms but the important part to me
is it would he says at the end it would
be very useful to see other people's
take on this proposal for Mortal
computation in the brain so please send
in your
comments but on the next slide just
before
discussion
um a few words of explanation of mortal
computation this was introduced by Jeff
Hinton in
2022 mainly in the context of neural net
learning and the paper I'm mainly
referring to is by Alex and Carl in 2023
where they relate mortal computation
particularly to markof blankets but also
to a lot of other things there are so
many ideas in that paper I I can't
possibly summarize them but basically
these are proposal for brain inspired my
biometic Computing I'll summarize I'll a
quote from that paper summarizes what
moral Computing means to Alex car
software is Immortal because it can be
copied to different hardware and still
be executable mortal computation means
that once the hardware medium upon which
a program is implemented fails or dies

the knowledge behavior and functionality
will cease to exist so that's model
computation um my take on it is that
direct physical computation which is the
kind of computation the wave does is
simpler and more efficient than a
universal Computing machine like a
universal touring machine or a universal
Von nman
computer and in particular Immortal
software needs very reproducible
Hardware but that doesn't work out in
the brain for two reasons two big
reasons firstly no two neurons are
identical in the brain I believe and
second secondly even one neuron never
does the same thing
twice so coming on to the relationship
between wave-based cognition and Mortal
computation I think wave based spatial
competition is mortal and it's morphic
to use a word from this paper in other
words it depends on the physical shape
of the thing holding a wave in terms of
their 5es model it is Elemental because
I think spatial cognition is a very
basic thing that any animal needs it's
embodied because embodied in space we're
all embodied in space and it is inactive
because we have to move in space in
order to understand 3D
space so wave computation is tied to a
visible wave and I will claim that we
need to understand 3D space before we
need to learn things and that's why way
spatial cognition is really very
Elemental and an interesting consequence
of this if you believe there's a wave in

the brain and if the the wave is the
source of Consciousness then you get the
conclusion that Consciousness is Mortal
I think we knew that but I think it's
quite
interesting so onto discussion and um
these are some of the points I would
like to hear about in discussion perhaps
you can take them in any order but um
Dan do you want to take
over thank you steering the discussion
yeah how about Alex if you'd like to
give a first comment and then anyone
else can can start to jump
in can you hear me okay let me know if
the background noise is really bad
because I'm in like a cafe so first of
all I think this is very interesting I
did read Robert uh your call for
commentary paper uh before as well I did
have a few notes I mean there's a lot of
things that we could talk about but I I
do really appreciate the connections to
mortal computation thesis uh which is a
corlar of the free energy principle so I
think this is really nice and uh I did I
did like the argument about that waves
are morphic thus you know that they fit
in the framework because we kind of need
that uh otherwise we're just dealing
with Immortal computation so I had
actually some questions I'll just throw
at you that might they might even just
be for further information so one one
comment question is I don't know if
you're aware of Robert West at uh Carl
tin University in Canada uh who has
worked on holographic declarative memory

uh one of your comments earlier about uh things being holographic and are there neur models that try to explain or try to model things that are wavelike but these aren't really wavelike uh holographic declarative memory comes into play where they use complex numbers and complex number representations which complex numbers are often used to model waves um so I was curious if you had known about that work or if you did uh why would that be insufficient or how that might play in the framework and I'll lump with this another body of works that way I don't break this up into too many questions uh resonator networks or resonating fire neurons have a wave like construction about them uh uh what is his name Eugene zovich in computational neuroscience proposed uh the zovich neuronal model which is a very complex spiking neural model and those Dynamics can be uh constructed such that they emulate resonate in fire Dynamics or wav like Dynamics and I was wondering how that if you were aware of that if not that might be very relevant in a potential neural model that could be a step in the right direction since you said that we don't really have the neuronal models or neural computational models that get at this notion of wave like comp computation um they might not be completely relevant so I was kind of curious about that I'll stop there because then I have like one or two other questions so I was curious if you had any comments on that thank thanks a

lot for that the quick answer is no I'm
not familiar with either of those and I
will go and look at them as soon as I
can uh they may well have insights uh
what I'd say about the wave hypothesis
is it
is the key thing about the wave and I
didn't emphasize this enough is it's got
to be a working memory so it's got to
hold information for a fraction of a
second
and that makes it rather different from
anything that is Neuron oriented but I
will certainly read those to with great
interest thanks a lot yeah I'll try to
uh send Daniel or you some links to some
of those papers to point you into those
directions and then I guess I'll um ask
my other question uh I mean I can also
lump in chrysiasis work on spawn because
there are lots of neuronal models that
try to get at modeling the Dynamics
through spikes and again maybe this goes
into that spikes just don't give us
enough Precision because in the paper I
didn't quite see that but in your talk I
did um which was why why do you claim
that we don't model spikes well as we
have a long history of spiking neuronal
models and computational Neuroscience
and brain inspired computing that do
work very well now again I don't want to
claim that there this is where Carl
could chime in as you mentioned him
earlier uh if there was any that have
been used to model aspects of complex
spatial cognition that I think might be
lacking but like we have hjin Huxley

model again I mentioned the isovic model
uh fiton nagumo there's a lot of these
complex Dynamics uh that can actually be
used to engage in very complex machine
intelligence tasks uh that for example
you can train simple robotic Control
Systems uh with like modulated forms of
Spike timing dependent plasticity which
is also a very rich and long history in
Neuroscience for plasticity of course
and this is where we're now we encoding
in synapses I'm not sure that this feels
at odds uh with the wave hypothesis but
rather could be tools uh that you could
be used to build wav like models and
again I was just curious to know if
there's still something about those
particular spiking neuronal dynamics of
which we have decades and Decades of
work that are just lacking to give you
what you seek um and again is it just
because we can't really represent things
with 10 spikes in five milliseconds that
even if we represent them Faithfully to
the neurophysiological data uh that's
still insufficient for representing what
you want and you're just saying we need
another mechanism that's the part where
I was a little confused yeah I I think I
please send the references I love to see
those papers I think specifically what
I'm saying is in I haven't seen a
detailed 3D model of space to the level
of resolution like a B's eyes or our
eyes that's actually built with neurons
firing there may be lots of uh neural
models built with Spike trains I'm not
aware of like I'm not aware of any

models within the Fe active inference
work I'm not aware any models outside
like Edmund rolls biset or um Jeff
Hawkins stuff and so on I believe that
all of those use precise theoretical
abstract
neurons okay that's fair um I think
uh yeah I think I I think I get what
you're trying to say I do I do feel like
there might be some interesting work out
there like for example I'm sure you've
heard of the recent model computational
models in the OS for like grid cells
which is used as an important part of oh
yeah no I think looking at grid cells is
an important thing we have to
do okay that might be complementary
rather than an opposition as I I do like
the way hypothesis I don't necessarily
think that this is data that would
necessarily refute it and this is where
Carl might chime in uh because again if
he's aware of any good neuronal models
or computational neuron models that
actually get at this that I'm not sure
that great
thanks Carl or Kenard jid feel
free perhaps I'll take the lead and just
um respond to those those comments I
think um you could argue both ways um
but it does rest very
sensitively on whether the wave is a
description of neuronal Dynamics under
the neuron
Doctrine or whether the wave hypothesis
commits to some divorce between the wave
however instantiated and the uh neuronal
activity so certainly

um the
the if you
say that the wave um is an appropriate
description of Ensemble
Dynamics um on the cortex or in the
hippocampus or in the
thalamus um then one could easily argue
that um it will be difficult to disprove
the wave of hypothesis and indeed there
are people's uh whose entire career is
built about uh built upon understanding
um spatial cognition and Beyond in terms
of cortical field theory for example so
we just look at the uh oh there are many
examples here
but the sort of U classical example that
comes to mind is the work of Peter
Robinson Australia um and developing
quite sophisticated wave likee models of
um neuronal Dynamics idealized can I
come in there car ju just briefly I mean
the key thing about this wave hypothesis
it's got be working memory so it's got
to store information for um and and it
is a separate non-neural wave that's
basically the way it's
pitched and it has to be a wave which
can store information for a fifth of a
second or a third of a second or
something like that and so it's not just
electric Fields passively caused by
neurons yes so that speaks to you know
what what does the the wave hypothesis
um commit to uh and you're giving a very
clear answer that uh it is not merely
articulating um Ensemble dynamics that
will could transcend neurons we could be
talking about G cells we could be

talking about Quantum coherence you know
but the the key thing is that you're
summarizing spatial extended Collective
dynamics of active matter in the brain
um yeah in terms of uh INF frequency
space basically so you could you could
say what this is exactly the same move
that quantum physics makes it's instead
of yes absolutely yeah um you basic I
mean at its simplest you can regard
quantum physics as basically
articulating Dynamics um be specific
about this sorry articulating density
probability density Dynamics in
frequency Space by appeal to complex
numbers and for transforms
um and um what's what's the benefit of
doing that well you can use foro
transforms and all the orthogonalization
and all the linearization of various
problems that ensues um what's the
difficulty well the slight problem with
um modeling probability waves is that
once you take a forer transform there is
no natural way to uh impose a su tow
constraint that you would need in order
to interpret as a Poss a probability so
what do you do you have to invoke a born
Rule and then think about probability
amplitudes but the message here is that
there is a great utility in taking any
arbitrary
Dynamics in some Metric space and
specifically density Dynamics and
projecting them into the space of waves
um and from that point of view you know
one could argue that this the wave
hypothesis is not a hypothesis it's

actually a principle or a proposal for a
better way of understanding Collective
Dynamics and in that sense I think there
are lots of examples of that and I come
back to cortical field theories Quantum
field theories of um um applied to um
sort of the brain um that have the right
kind of damic will have exactly the
right kind of um um working memory so
I'll just give you a couple of examples
there is work um looking at um Chimera
States in cortical field Theory um where
um one can find things like solons which
are you know brings us back to last
year's conversation condensates and the
like more interestingly more recently um
there's been empirical evidence from
both EEG and
fmri of vortices on the cortex um which
are if you you know one way of
understanding um classical like Dynamics
um in the space of waves that have the
kind of
itinerancy and
um solenoidal aspects that do remember
things by returning to the same state um
so if you know I think there's lots of
empirical evidence U that and thereby um
meaning that they have memory in the
sense of say a konari map um and these
are explicitly articulated in terms of
traveling waves that have this circular
noal aspect that can be empirically
detected however so I you I think your
answer the question is there any
empirical evidence for a formulation of
um brain Dynamics Beyond um a commitment
to particular the neuron Doctrine um

that can be articulated in the space of
waves that fulfills the the uh
requirements of precision for memory and
representation I would say yes but that
does not answer the question
is are these waves mere descriptions or
of one or many kinds of descriptions of
some
underlying um some underlying
Dynamics um or does the spatial wave uh
the wave hypothesis of the brain say
that the actual substrate the Mortal
substrate cannot be just limited to uh
message passing among uh among neurons
and that's what I meant by the neuron
doctrine that I think is
yeah yeah that I think that is my view
it cannot be reduced to neurons
basically yeah that's what I understood
from your previous talk in the slide you
shared before of course there is there
are a lot of waves in the brain for
instance related to electromagnetic
induction and there are a lot of old and
very powerful models go back to nun
years you know to connect with SCS
discussion of EG Dynamics but I think
one of your points is that the wave
whatever that is uh cannot be just an
epip phenomenon and it has to have an
inductive power on neuronal Dynamics in
return um there is some model I think
that are emerging uh or some theories
that are emerging about electromagnetic
induction uh uh like electromagnetic
wave in the brain that with inductive
power on current in the in thees um I
haven't followed that literature for a

while as I told you before I give up
gave up on Neuroscience uh but um so
what you require is really something
that has a you know bottom up and top
down quote unquote sort of mechanism to
begin with this wave has to be sustained
and it has to be in in of itself a form
of computation that can then feed back
to the noral network that's
question yeah
Ken would you like to give any
comments uh just to maybe a one for
everybody a question for everybody might
it reveal some of my ignorance here but
um is are there any sort of General
results about the capacities of waves
for computation relative to other types
of computation it's it's a kind of
analog in computation isn't it it is
analog is very appropriate term for it
but I I would say the the role of the
wave is more information storage than
computation but it it actually may be
involved in both but the key thing is
it's got to neurons May compute 3D space
there got to be some way of storing it
with adequate precision and that's what
the wave is supposed to
do but I should add if I may that
Computing 3D space in particular in an
active in C kind of a realistic
biological uh context for you know
animals that's not just representing
space it's also acting upon it through
Transformations that are not just about
change of coordinate for instance under
hypothesis there is the projective uh
group that is acting on a projective

space but then you have the space that
is itself filled with features that
themsel as their own operators like
different groups for and spaces colors
and how all these things like perhaps
through models of fiber
bundle interact is very key so the the
computation of um space as you think of
it and I think is perfectly correct and
and talks to me uh is much more than
just uh you know represented
representing three coordinate at one
point in time I agree I
agree just to speak to um Ken's question
about are there any laws that would be
applicable under the wave
hypothesis just going back to the
analogy with quantum mechanics if one
thinks of the wave as a
probabilistic um representation or
memory um then you'd certainly be
looking to Heisenberg and certainty
principle and the de brogly
relationships so I perhaps this could be
a question for Robert you know H how
would you unpack the spatial Precision
in relation to the temporal Precision
because you talk about memory but the
the uncertainty principle would mean
that you can um you can have a certain
um spatial Precision at the expense of a
temporal precision and vice versa indeed
that is one if you like gift of a wave
hypotheses certainly as read from the
point of view of Elementary uh qu Quant
quantum theory um the the other so
question i' reflect to you in terms of
the deoy relationship what what what

what is the analogy of say energy and
mass in in these in these ways you know
how would you in how would you interpret
this to a sort of 101 physics student um
when it comes to with great difficulty
um I let me first speak to the spatial
Precision I think the uncertainty
principle is basically a property of the
fer transform a wave that's confined in
a certain distance has a certain
uncertainty in its wavelength and
wavelength is momentum in quantum
mechanics and and position is position
so that's the UN that's the the origin
of Heisenberg uncertainty principle is
simply the Freer transform of a bounded
wave that speaks to the spatial
Precision but as to the actual
biophysics of the wave and there is some
relationship between wave vector and
frequency I don't know what that
relationship is but the the gr as you
recall
uh which way does it go frequency over
wavelength is Phase velocity and the
derivative of
wavelength with respect to frequency is
the group velocity now all those
questions about what the group velocity
is what the rest mass of these if there
are Quasi particles what's their rest
Mass I don't know but uh for instance
one of the more far out hypotheses I
consider is as we discussed it last time
was that the W is actually held in a
Bose Einstein condensate and in a Bose
Einstein condensate there are particles
called roons that haven't phonons

normally have zero rest Mass but there
are particles uh called Rons and others
which have nonzero rest M and that may
be something to do with their
persistence in time but nonzero rest
Mass gets you into the whole question of
what's the frequencies of this wave what
are the wave vectors
and as far as I'm concerned it's all the
Wild West it's all to play for really I
I I'm only making suggestions at the
moment thanks I'll make a few short
comments so thank you all for the
discussion and Robert also for making
the basian epistemology very clear I
think it's really cool also that was
very insightful discussion on one
possibility the wave is epip phenomenal
like David said plays more of like a
memory possibly even a memory with no
causal
efficacy no I'm sorry I did not mean
that I mean according to what I
understand and is necessary
for Rob model the wave must not be a p
phenomenal yeah and that's right what we
know about EG except some models for
instance now there's some models of
electromagnetic propagation with notion
of feedback induction that should be
looked into but overall the way we model
EG for instance from the neuronal
current is EP phenomenal it's just you
know summation at the level of an
electrode and so the wave itself is an
epiphenomenon and the the model of
Robert does not want that I totally that
agree and that that's where I was going

was sorry to attribute it to you that
way but it was yes to to a more active
computational role kind of like this
Reservoir
Computing area um also a general Point
kind of summed up by figure 4.3 in the
2022 textbook is there's like continuous
and there's discret models in space and
in time
so given the continuity of space and
time just at a first pass a continuous
representation or model like a wave may
have certain advantages over holding it
in a discret way and then um in the act
of inference does not contradict folks
ology work there's the continuous time
model on the lower scale and then a
discrete more switching slower model
that could be like an intentional
context oh so so there is precedent in
terms of having a lower level continuous
model higher level discreet models and
uh then more to taking that to the
bigger
questions there's also some natural
empirical correlations with bigger
slower things and faster smaller things
so to the discussion of well multiple
things are represented like space and
time some of that is doesn't need to be
an open-ended representation it can be
like evolutionarily tuned or
developmentally tuned or learned um and
then last comment just on the insects it
was a great point a a line of evidence
that's very promising because there's a
lot of neurophysiology and neuroanatomy
available in insects and just to say

that the
visual region of the the brain this is
arthropods and insects and the old
factory as well as the mushroom body
have more like stereotypical units and
they do have more change in size and
more variation in evolutionary whereas
some of the interesting evidence from
what you presented from the insect brain
is that the actual geometric volume and
size of the central body is relatively
conserved even as other parts of the
brain have a lot of allometry so that
suggests that one way or another either
it doesn't need to be explained in terms
of fulfilling a adaptive function it
could just be oh it's contrain or
something like this but if it is going
to be playing a a role which is a fair
assumption in an energy limited system
then part of its role can't necessarily
be improved by making it bigger or
smaller yeah I agree I
agree you mean relatively bigger or
smaller right because I mean it the
central body does scale a little bit
with with the insect brain size but it
it scales less rapidly than the total
insect brain
size the interesting to hypothesis see
whether it scales with the visual Acuity
of the
insect yeah how the different regions
size and other attributes this could be
in mammal Andor arthropods power the
empirical psychophysics and the just
noticeable
differences described or explained by

like for example just geometric volume
of the old factory L whereas if the or
is there some other variance to be
explained
yeah well but there will be a lot of
correlations between you know neuronal
population densities and and volume so
if volume increase you may have more
neurons
uh obviously you know that's basic
neuros psychology 101 and vision
Neuroscience one1 you know if you lose
the neurons you acity diminishes so
there seems to be a strong correlation
between your angular Precision or acity
for many other things and the number of
of neurons and connections that those
neuron makes but after a while if you
lose twoo many neurons you have a de
differentiations like think about neurod
degenerative diseases like you know take
alimer for instance where everything get
disconnected and there is not enough
neurons to represent stuff so that I it
would be hard in the C you know given
what is fairly know in Neuroscience to
reject the importance of the I'm
simplifying here the number of neurons
the number of neuronal units in your
brain and your uh resolutions capacities
thank you yeah how about Robert continue
with uh part two should I I bash on yeah
great well thanks everybody for
that uh so here's what we've got to on
the road map we've described the
hypothesis evidence for thre against
we've done the assessment and we're now
on to Future tests how can how can we

test this way hypothesis and I'll divide
this roughly into what can we tell from
aonomus from the insect central body
from computational models from
biophysics and from other sources of
evidence so I'll just go through some of
those um so um this probability 0.5 is
an unsatisfactory place to be we really
want to drive the probability up or down
and so the first thing to do is to
stress test the three main groups of
evidence that we have already and then
we look at other evidence for and
against and there are a lot of possible
tests described in this paper which I've
given the link to the P tests are all um
in this table here now I'm not going to
go through all the table uh I'll just
say the paper allows you to sample the
particular area that may interest you so
you can just pick out the section that
speaks to whatever test you're most
interested in and I'll just do some
dipping in sampling
myself so the first first thing is the
so-called What I Call The Smoking Gun
the specific evidence in the thalamus
and what this specific ele evidence is
is there's a picture of Thalamus here on
the right and so Thalamus is a round
thing at the center of the cortex which
is like a hemisphere around it and what
you have to notice is the thalamus in
mammals has lots of different nuclei
thalamic
nuclei and the thalamic nuclei are only
very weakly connected to each other
but they are strongly connected both

ways to the cortex so the question is
why don't the thalamic nuclei move out
of the thalamus towards the cortex
they're going away from each other there
only weak connection with other nuclei
they're getting closer to the cortex so
they're reducing the length of mated
axons and that would save a lot of
energy so that's the problem with athus
under the nuclear model if the nuclei
exists just to have neural connections
to each other they're very weak neural
connections and energy could be saved by
moving out each nucleus towards the
cortex whereas if the thus has a wave in
it there's a good reason for all the
nuclei clustering together the nuclei
have to be together in the same
wave now I've tested this energy
argument by
actually uh modeling what happens when
you move theic nuclear
and approximately I obviously it was
very approximate test on the human
Thalamus and the answer was yes you can
save energy that way so there is a
mystery of why the phalus stays together
but that was an approximate test I think
now that test could be done much more
precisely using connectone data brain
connectone data so you could use brain
connectone data you could have a
computer program that tries out various
moves of nuclei to see if they save
energy so that that is the test that you
could make this argument about the
Smoking Gun much more
precise and there's a similar test for

the thalamic particular nucleus which I
won't go into I would
say sorry sorry to interrupt I would
have kind of is it possible to interact
Slide by slide or do you prefer that we
yeah absolutely
yeah so I I get your uh energy argument
in term of neuro development or neuro
evolution of anatomy and you know
anatomical relationships though I would
argue I mean I would not argue I would
you know ask you but you know what if in
this uh energy model you take into
consideration all the inputs and the
outputs uh from Sensory neurons for
instance from sensory organs the eyes
for through the optic nerve uh and of
course then the output toward the cortex
but also the you know motor input uh I'm
not sure uh that if you think about all
this wiring then your energy model not
forgetting all the white matter that you
have you know and connectivity that you
have between the cortex and the thalamus
not sure that the energy model would
fair that well that's the course um well
the basic point is that when I did the
approximate calculations I did
take account of the
upcoming you know the sensory neurons
coming into Thalamus the point is if the
phalam was is just a relay pushing data
on from sens dat to Cortex then moving
the thalamic nuclei would be neutral
whereas because there's feedback
camic connections you actually gain an
energy by moving out towards cortex so
this calculation has been done albe it

very proximately and I think the
opportunity is to do it much more
precisely taking account of all thalamic
connections from connectome
data uh but you you know of course and
that's really a bit uh old for me I
haven't been into this literature for a
long time but there are B models
connectivity it's yeah thalamic nuclei
kind of lives in CEO but they actually
have Silo I don't know you know there
mod but they actually have bodal or
triod uh uh interactions that are kind
of fine some of them are modulated by
the reticular nucleus is that with each
other I understood from maray Sherman
there's a quote from him that says there
are no connections he puts it very
strongly when was that published remind
me um well I mean he publishes book
about Thalamus in 2006 but yeah since
then he's written a lot of papers about
Thalamus and and he's
well yeah sometime people are very
assertive but you have a great all that
neuroanatomist that I've met in the past
like G vanen uh uh they look at the
brain in very very very very small in
minute details you know they they do
real microscopic Anatomy and the way
they see the brain universe is quite
different from uh I don't know if this
nor anatomist you are talking
about work with these methods uh I would
be
very prudent with connectomic
data I think they are useless but anyway
that's another so you suspicious of

connect home data yeah hold the DWI I
mean DTI data and function okay that's
my personal opinion but I'm radical I'm
almost you know I think they are useless
lot I think they are useless but anyway
we can have another discussion about
that
yeah anyway to me there's quite a close
analogy between this argument about the
thalamus staying together and the
argument from Galaxies for dark matter
in other words when people looked at
galaxies a long time ago they found
there wasn't enough mass in stars to
hold the Galaxy together by its gravity
so they invented Dark Matter so I think
when we look at Thalamus there's not
enough neural Connections in the
thalamus to hold it together against the
pull outwards towards the cortex and and
so the wave is a bit like dark matter in
the thalamus
so that's the analogy anyway but I think
the connectome test should be a precise
way of uh stress testing this
hypothesis right the in central body um
as I said the gross anatomy suggests a
wave but you can't make the same Smoking
Gun argument but I think there's a lot
of detailed Anatomy that you look at of
detailed neural connectivity where the
synapses are distributed and where what
happens in the compound synapses I think
they're very
suggestive and there may be not a simple
connectone test using nuclei because I
don't think there are nuclei in the in
central body but there may be some

connectone test
that can reveal whether the Neuron model
actually is sufficient to account for
the shape of the central body so this I
think is a very open area for
investigation I personally like it
because I think science should be done
on simple things and there's this quote
from Ramon kahal that said the melan
brain is like a grandfather clot the
insect brain is like a Swiss watch I
like that so I think we should look at
them so the Third
argument we should stress test is neural
computational
models uh I make this claim that
understanding 3D space is very Elemental
it comes before learning I think because
animals have to do that before they can
start
learning and I've built a model of 3D
spatial cognition at level two and that
needs High precision and had this
argument before new or Spike
trains only give much less Precision in
short time so what we need to do I think
is to stress test this argument by
really trying to build a model of a
neural model of spatial computation
using mortal fible neural neurons and
one good framework to do this is is the
free energy principle because it's got
well established neural process model
and if we can make this work with
neurons then maybe a wave is not
necessary but if we can't make it work
with neurons then that that strengthens
the case for the wave just another slide

on this I think we should try and build
two models say in the F
Fe um perhaps using the level two model
as a starting
point but in the moral neural
model we have to model do we model Spike
trains or do we model just a variable
neural Precision there must be some
neural in prision we model if we model
neural in Precision how do we do
gradient descent because you'll see I've
struck out here if you do naive
calculation of gradients in other words
you compute a quantity and then you
shift some input variable small amount
and you compute the quantity again to
compute the
derivative if there are lots of Errors
you don't that doesn't work because the
imprecision overwhelms the
gradient so there's an issue there about
how you do gradient calculation I'm not
sure
how you tackle that obviously you can
compute gradients explicitly but it's a
more complex computation than Computing
a
quantity next question how do you do a
gradient descent in
space and I think in the FP a gradient
of a free energy is a three vector and
Precision in the F sense is a tensor
probably how do you what coordinate
system do you use do you use cartisian
coordinates I don't know I mean seems to
me cartisian coordinates in the brain is
a bit
unrealistic so the key question I think

is I expect that the naive Precision of
neurons from Spike trains is not going
to do the job so the question is can we
have some clever neural Micro Design
which gives us High precision and high
speed and the final question is do we
use a spatial hierarchy as is common in
the F or do we use a Blackboard
uh again I would prefer using both but
the issue of what are the mark of
blankets arises there so that's the
issues we' have in a pure neural
model in a wave neural model how do we
model a wave do we build a physical wave
now that suggestion is not quite so
stupid um one could build it um in the
very early days of computing there was a
thing called a Mercury delay line memory
which actually used a way for storage in
one dimension and we could build
something like in three dimensions but I
think that would be very expensive not
as expensive as building a quantum
computer but very expensive anyway so do
we simulate a physical wave or do we
just simulate the memory parameters that
we get so simulate the Precision and the
capacity and so on I would prefer to
start with a simple simulating of the
memory parameters and in other words I'd
like to move down from level two to
level three in successive
stages so that's um the Active Vision in
Exploration another area we can look at
is if there is a wave in the brain it
should show up in genomics and
proteomics
somewhere and there should be some genes

which express only in certain places
like in the thalamus or in the insect
central
body or those genes May the proteins may
be particularly affected by anesthetics
because
on the hypothesis that anesthesia
disables spatial cognition disables a
wave you might expect a a link between
those proteins and anesthetic
molecules and we're looking for proteins
which not only sustain the wave but also
proteins involved in neural transmitters
and receivers which exchange information
with
neurons and I
think that's a promising line of
Investigation if we find any possible
proteins they could help to constrain
the biophysics which I think comes
next biophysics of the brain is
absolutely wide open I have no idea
really there are some very preliminary
Clues and I think most of the ideas are
probably wrong but the key thing is the
wave needs to store information it must
persist for fractions of a second so
it's probably not purely
electromagnetic uh because as David said
that's an EP epiphenomenon activity so
could it be a coherent Quantum
effect uh the problem about quantum
mechanics of course is
decoherence removes it very quickly and
you have to somehow conquer
decoherence there are ways to combat
decoherence I'm particularly inclined to
interested in looking at nuclear spins

there's a chap called Fisher who's
looked at spins of the phosphorus
molecule which he claims can have
coherence times as long as seconds or
even minutes
um there is
this possibly attractive idea that the
wave could be in a Bose Einstein
condensate or a Fric
condensate um so there all these
questions all these possibilities I
haven't really any strong direction or
other but we have to answer the question
a how does the wave persist and B how do
neurons couple to it and we might get a
clue in that respect from single celled
animals we'll come to that in the next
slide
but I'm saying this is a tremendously
difficult problem and just because we
can't think of an answer we shouldn't
underestimate
Evolution and I think broadly we need to
find some possible me before we can
start looking for it
experimentally
um this is intriguing to me um there are
single- celled animals like eroid is I
can't pronounce it it's a single cell
animal it's kind of vanav and it has
complex Behavior it hunts and eats
crustation eggs and it has something in
it which looks suspiciously like a lens
of an eye and something which might well
be a retina so does it have a lens and a
retina if so how does it use that
information if it hasn't got any neurons
how does it guide its own

behavior can there be a brain without
any neurons and could that be a
precursor for all the brains could it
arrive evolv in prean Era and and
persisted through the caman era so I
think there is something to be learned
by looking at these single cell
animals may I comment again yeah sure I
totally agree with you if you look at
paramia obviously they are very
sophisticated sensory motor schemes they
have some there's some evidence that
they can have some memory they have can
be conditioned they have potentiation
some of them like spir spirostomum if I
remember well and obviously the
computation quote unquote or the analog
analog computation if you prefer is
going through molecular networks but
works exactly as well as a as a
collection of
neural animals uh so that's also is
telling that computation again quote
unquote memory and active inference uh
can go through many type of biological
networks uh and so we shouldn't think
that because the neurons are the most
appearent and they are very very
sophisticated and complex that uh there
are no other kind of unknown for us
mechanism of computations that go
through biological network of protein
network if you wish that I think it's a
very interesting hypothesis and that the
multiscale interaction between all those
things is also kind of an entire
universe that we you know barely
understand mro

gal cells are now we understand better
that they have not just a trophic hole
but a very sophisticated computational
role so anyway uh I think that's that's
a very important thing to consider in
know for the future of
Neuroscience great thanks David yeah so
I've really finished my talk um over to
discussion again um so some of the
topics we might touch on with the
connecto ISS
or particularly the computational model
very interested to hear Carl and Alex on
that and insect brain studies and even
the biophysics so um over to you
please
naughty um is preparing his um revealing
answers and commentary I'll talk rubbish
for a moment um so I I just pick up on
the sort of practical issue of the
connect term I think I I think um David
was showing an understandable allergic
um response to connector based upon
diffusion weighted Imaging using
non-invasive magnetic resonance imaging
um I think what you're talking about is
this connectivity that could be tracing
studies it could be um supplemented with
problemistic trap toogy but you know so
if you just don't use the word connector
I think I think um uh you'll be fine
just talk about connectivity studies um
which are usually definitively um based
upon sort of you know staining and
tracing um at a microscopic uh level as
opposed to right
reconstructions um for me um well you
know the tests of the hypothesis

um there for me the ones that you've
offered um they do um invite rather um
if like deflation responses so for
example the arguments that a single
neuron firing um over a 256 millisecond
um period cannot um contain the
sufficient statistics required re ired
for spatial temporal Precision of the of
the sort required for um spatial
navigation at a certain space-time
scale you know one simple way of
counting that well that's true but if
you give me 10 to the three of these
spiking neurons and I can now just write
down um the um Ensemble average then
they do um so the whole Precision
argument can be I think almost dissolved
without making any iral moves or any
tests can I come in on that I I think if
the test You' you've had your chance
fromer no no no this is
no just briefly just briefly the test of
the Precision argument is to build a
model that works and by the way do it
for an
insect yeah um so I think we're getting
pretty close to that but and in so doing
um you actually appeal explicitly to
mean field approximation which is
exactly modeling
the probability distribution of
populations or ensembles in the S
mechanics sense so you know you know the
I I'm reminded now of um Rodney
Douglas's ambition for the EU to build a
b uh I think if you look at certain um
drone rumber applications of Slam
simultaneous localization and mapping in

Industry actually out there um you will
find that the you know not perhaps to
the level of a bumblebee but certainly
to sufficient to um clean up the airport
floor amidst moving uh passengers that
kind of technology is already there um
and you know I would imagine in about 3
years you will have um the same kind of
Technology um operating at the level of
drones and and at that point I think you
can say yeah we can do this without
without a wave hypothesis um yeah but
drones with respect have perfect
theoretical accurate neurons they don't
have mortal
neurons that's yeah the point being that
you can simulate a population of mortal
neurons such that the random uncertainty
averages to zero so this is what is
meant by mean field approximation so
that all of stochastic and Cal
thermodynamics rests upon this so the
density Dynamics itself now becomes
deterministic you are still modeling
something which can be quite noisy with
lots of random fluctuations and
thermodynamics underneath them but the
density dynamics of the kind that
quantum physics would appeal to is
actually deterministic but I'm not I'm
not you know reason I sort of was
teasing you said you can't respond
because it's our turn now was I don't
think I don't think these are important
arguments to me the important argument
is something which came out in the last
discussion and you I think Daniel
summarize very nicely is is there any

top down causation between the wave and the neurons that are controlling my body and my eyes um and um you know because I could argue that it's all bottom up causation and just rewriting neuronal density Dynamics or population Dynamics in Foria space yeah everybody does that even to the extent of um focusing on the relevant variables as the rate of change of phase which Robert you were you saying that a really interesting interpretation yeah that is the kuramoto model of neurodynamics uh you know this has been around for decades people do work in this space but the assumption is that you are basically appealing to a bottomup causation from some underlying neur mortal computation at level of neurons and then you're just lifting it to a level of description in the in in a in a poror space or in a phase space with the right kind of Dynamics making all the arguments that you have made in favor of the wave hypothesis it seems to me the important thing is a top down causation that destroys theological thing so I'm just trying to think about tests of top down causality of course not easy to do but I'm just wondering you know if I squash your Thalamus would you have different spatial cognition so I preserve the connectivity but I actually changed the shape of your Thalamus things like that I think would be more compelling demonstrations of the top down causation which I think is at the heart of your spatial wave hypothesis that it's not just an

epiphenomenal bottomup consequence it
actually matters this wave my neurons
have to be immersed in this wave to
function properly and it matters where
they are and in that mortal sense it
matters um that these anatomical
constraints afforded that support this
wave will have an influence on
neurodynamics and subsequent belief
updating and movements and sense making
and projective geometries um so I don't
don't have an answer but I would love
look into you at at those things that if
you like address the notion of circular
causality from the point of view of
physics and in particular drill down on
what would it be mean to demonstrate
empirically top- down causation in the
sense or the spirit of George Ellis um
that that's what I took from that
conversation that exchange I I agree
hard
problem honestly I think even though
Carl you said you're just gonna you know
Babble or say rubbish I think that was
the most important point because that
touches on the things that we even
discussed in the in the Mortal
computation thesis framework um and that
was actually so I don't know if I have
the wonderful insightful answer that you
claimed I might produce but uh just to
speak to what you said and uh and what
Robert was discussing I think that is
what were lies the most important if
we're going to build computational
models which obviously appeal to me
quite a bit

uh in the moral computation thesis said
well okay the real way to do this is we
would take actual substrate like for
example a neuromorphic chip if we were
doing
artificial or we would work with
xenobots or anthr robots and build you
know little systems of these actual
biotic or chimeric type of systems but
we did discuss virtual morphology so I
think you could just go down to the
simulation route which touches on what
you said Robert about well we could
build waves we have examples of those
but those come with some difficulties we
could just simulate them and I think the
answer to answering the question how do
neurons couple to the waves and what
does that look like and how do we build
those computational might be one of the
most important questions I think in
building any testable computational
model and any useful simulation I do
think probably what I was saying in the
earlier discussion about looking at
resonator neurons or resonating Dynamics
uh which has some history and as well as
the cognitive math learner and the stuff
I've talk about holographic memory which
kind of Builds on the complex number
representation and the wave
representation and how do you encode
that in spiking Dynamics I think that
might be an interesting and useful
building block and I will send Daniel
and you some information and then you
could look and see if that's rubbish or
if that might be useful but I think that

might might be important building blocks
to actually building a workable
computational model I do think
simulating the wave could be interesting
to from a different direction I was
another thought that you produced was
from nature inspired metah huris
computation or optimization where we
model things like bumblebees and
beehives and ants and ant colonies and I
know Daniel has done work with ants as
well and likes them quite a bit um and
also we've done things with bats
echolocation and wave transmission so if
we model mathematical waves at least we
could use that or Frameworks for metah
heuristic computation and sort of
backwards build that let's say derive
the computational mechanisms that could
allow us to couple neurons to use that
information and then we could do actual
simulations of different wavs of
different properties we can model the
frequencies we can kind of control those
hyper parameters and then in doing so we
could then look at
what Carl was then bringing up which is
the one of the key Essences of mortal
computation which is that spatial
Arrangement that anatomical organization
of uh neurons and their particular
components um then you could also do
tests like okay if we build them in a
spherical kind of construct or we build
them in such an anatomical construct
that um is conducive to those simulated
waves then you could do what Carl said
computationally which is we could change

the the shape of these anatom these
computational Thalamus models right we
can make them cubic or you know other
types of weird shapes and see if
squashing them changes for example how
these neurons respond on simple
cognitive tasks that we can emulate
again we could use very simple spatial
cognition tasks that we could simulate
and then we could start manipulating the
actual uh virtual morphology as Coler
would call it or just the virtual
anatomy and then that could give you
some at least computational tests of the
effect in value of wave computation um I
think that's the key and I'm not even
sure if what I just said mostly is
rubbish and uh just me kind of
generating what I would do if I had to
build something like that and I think
that might be one of the single most
important questions because I think it
answers both neuron models without waves
and neurons plus waves as you said we
need to build both those models and I
think that kind of coupling and that
modeling of that relationship is a first
step and then of course then you could
then start to look at actual physical
constructs that ulate that computation
kind of like a reverse we're using
computation to drive what we think the
structure should be and then say are
there actual structures that we find and
then we can constrain our models a
little bit better as we do have anatomic
or we do have computational models of
phalic structures like Chris alus group

has a phalus structure in their nango
framework uh again spawn I think has
like a Thalamus Arch component um so the
and those are very biophysically
grounded like they use anatomical values
things that are actually grounded from
neurophysiological data so that could be
an interesting starting point and that
could allow us to then actually test and
uh simulate what Carl was saying about
squishing and stretching and uh changing
the form of the thalamus so that's
something I thought might be worth
mentioning and that's where the Mortal
computation theis really comes in play
because again it's that specific
morphological Arrangement that actual
construction and relationship between
your owns that's really important and
changing that would change the
computation that's what another
byproduct of our framework was basically
saying is that you're not going to get
the same computation if you change the
anatomy if you change this um so that's
that's an important part and then just
quickly I wanted your last slide before
this one I don't know how to pronounce
the name of that uh that entity either
um but I thought that was really
interesting because it you were saying
it has essentially like a little eye but
it doesn't have any known neurons or
anything that we can perceive directly
so could that be neurons without
cognition and I think that gets at to
what uh uh Michael Levan and again Carl
and I site and talk with him a lot which

is the idea of Basil cognition and I
think that's kind of where we don't
really have that quite right from a
computational point of view so I do
think that's even more important to try
to model these uh Elementary cognitive
systems and again looking at these
building computational models I think of
like that single cell organism could be
valuable and maybe give us some insights
as to well okay what is it that we need
to get this thing to actually solve very
very simple tests like follow gradients
of chemicals which we do that in
chemotaxis and phototaxis experiments
and we do build little models uh of for
example Eola trying to Traverse these
little uh little in niches and I think
that's the other part that you're going
to need in building uh even simulated
versions of what you seek which is
you're going to need that virtual Niche
because again and you point out Robert
which I was happy to see that that you
know your what you're thinking is that
it's Elemental embodied but I think the
other part that can't be ignored if
you're going to properly uh examine and
simulate mortal Computing systems is you
need a niche need that more uh virtual
environment to actually and those
physical properties need to be very
explicit um when you build it so I think
that that was something that I think is
going to be also very difficult uh but
probably the most important to make the
next step for Mortal computation and bi
intelligence hopefully some of that made

sense no thanks Alex I I agree with you
that the simple animals the simple comp
simple brains are the place to look I I
was recall that quantum mechanics
started with the hydrogen atom not with
the uranium
atom I'll I'll actually kind of add a
complimentary view I was wondering to
what extent will the kind of
representationalist form and context for
the wave that you described in humans be
of maybe even a slightly different kind
in a single cellular or in an insect so
for example there's the outfielder
catching a Fly ball who yes you might be
able to describe that in terms of the
spatial temporal frequency of the
baseball moving through space and the
body moving through space there also may
be a horis such as following a gradient
of shade that also gives a behavioral
solution or even an indistinguishable
approach so th that may speak to again
the variety of possible substrates of
mortal computers because the insects
central
body and this is the to the empirical
Point how can we differentiate the
situation where spatial representation
is happening with or without a causality
closure from where behavioral strategies
are integrated like moving away from a
shadow for a fly it may not require the
spatial Precision on the object so
actually just determining where and how
variation across different species and
with across individuals and settings how
would we differentiate that from a null

hypothesis where neural or wave wave
particle or both or neither we don't
even need to invoke space we just invoke
reflexive
Behavior I agree that's a very important
question it's addressed a little bit in
the paper but what we need is tests of
insect behavior for instance which
depend on the 3D spatial model which
don't depend on just as you say
detecting a shadow or detecting a
looming thing uh and it's quite
difficult to devise those
tests yeah I'll read a question from the
live chat and anyone can can give a
thought okay Courtney writes thank you
for the interesting talk could this
Theory be used to test whether or not
someone in the ICU with brain damage is
conscious and if so to what degree could
it be tested similarly to integrated
information Theory or are we far away
from something like
that I'd say we're far away because even
experimentally finding the wave
is far away we have to work out what the
biophysics of of it is before we can
build detectors for it and that may be
as far away as as detection of dark
matter or for the neutrino for instance
that was theoretically predicted it took
25 years to find it experiment I think
it may take a long time to find a wave
experimentally so in terms of measures
of Consciousness I'd say we're a long
way
off thanks um Ken deid any please go for
it well I just wanted to mention one

thing um that this whole discussion
reminded me of a long time ago I think I
was in graduate school I read a
book by Ralph D Ellis not to be confused
with the George Ellis that um that Carl
mentioned but it's called a book called
an ontology of Consciousness and if if
my memory serves it defends just the
general idea that Consciousness is a
wave um and defends the kind of w wave
medium model for thinking about the Mind
Body problem and I I just remember it
being very interesting but I don't but
that's so long ago that you might read
it I might read it now and be totally
disappointed but just something to throw
out there send me the reference that be
great yeah yeah Ralph DL I can send you
that an ontology of Consciousness but I
do remember being being struck by it as
being sort of plausible but you know how
those things go it's like it's
metaphysics it's
ontology you know yeah you're pretty far
away from physical testing but in terms
of some of the sort of theoretical
advantages of thinking about it that way
it might be worth a
look I'll put it in the chat thank you
de any thoughts and then anyone watching
can write any last questions and then
otherwise Robert let's hear like what
are your next moves and just kind of
summarize the commentary
affordance maybe I I can you know I
I this is a kind of question that you
know anybody interested in topic
struggle with in kind of okay what is

the sort
of Integrative mechanism uh that is
required to understand things like
Consciousness uh or does it relate to
the brain with all the limitation of
both you know theoretically and
methodologically of of contemporary
Neuroscience uh I'm totally open now at
this point to the possibilities that
yeah there is you know some form of
physics that we haven't understood about
the brain or that is related to the
brain or interact with the brain and
becoming like even open to some form of
you know physical dualism I don't know
you would call that uh I'm sure that
it's it's going to be very hard to
progress with that that we are still at
a sort of metaphysical age of this type
of Science and discoveries uh we don't
have the principle yet maybe uh I doubt
but I'm not a physicist so it's really
like you know talking that Einstein
condensate will address the question I
mean think about the the size that has
been achieve the maximal size of both
Einstein condensate and under what
circumstances irrespective of you know
okay uh temperature issue you know but
it's what like a picol lit like billion
atoms um so you would have like billions
and billions of bosen shin condensate
that would have I mean it's it seems as
I I don't I I don't have a good gut
feeling about it
but uh there is also more mundane things
like you know we had this discussion
about uh electromagnetic wave in the

brain at the kind of micro scale or
mesoscale as EP phenomenon but as I said
before although I don't recall the names
of the authors working on that but there
is a bunch of labs and researcher
working on that uh there is now or at
least there's some theoretical well
formulated ated theoretical ground and
some empirical evidence for inductive
mechanism in the brain basically you
have you know currents going through uh
um your axon bundles of fibers uh you
know your action potential if you wish
that are associated with those currents
that induce electromagnetic wave that
propagate and may have a role of for
instance synchronization with another
bundle just next to it so are those
things are very minute as effects
apparently uh they don't have a very
very large long range uh um effect
that's because of biophysical constraint
if I recall well but it talks about the
possibility that you know we may not
want to throw the baby out of the bath
water classical electrom magnetism may
still have something to say and in
particular if this kind of induction
mechanism uh which is literally top down
I mean the downward cation if you wish
you know there is a fiber bundle that
generate electromagnetic waves that then
affect currents in another fiber bundle
we could say well you know then you
could imagine something quite
sophisticated and in which the role of
anatomy the grow Anatomy the shape of
the brain the organization then might

become uh very important to understand
because of course the relationships if
those electromagnetic wav induction are
short range you know how you bring
anatomically things closer in order to
make them interact becomes very
important and maybe that connects back
to your thalamic hypothesis but again
what I'm talking about is very very very
speculative thanks
yeah I agree um my my feeling is the
biophysics is wide open so
electromagnetism May well be a part of
it and you used Mars levels earlier it's
possible the wave is has different
implementations like I think this
conversation brought up a lot of
different
dimensions and also even begets bigger
questions
like do scientific hypotheses have for
and against is there only one hypothesis
chosen or could there be
functionality distributed across
multiple biophysical systems such that
something being a AR true or false which
is kind of predicated the whole basian
epistemology that doesn't necessarily
need to be true there could be utility
continua for a theory and a predictive
continua yeah I I
guess the hint is that since there's
ever a hint of the wave in both the
thalamus and insects and opposite ends
of the animal tree evolutionary tree
that the wave hypothesis might carry
across the whole lot and in which case
you ask is it convergent evolution and

different species different branches or
did it go right back to the pre
primordial Branch before the Cambrian
era I would like it to go right back
yeah in that case you can consider like
arthropods or look how far back but
insects are only one kind of arthropod
and they have yeah enough homology that
you might be able to extend this
argument
and do ancestral State
reconstruction on neuroanatomy at
different levels and these analyses may
have already been done and then the
allometry of those reconstructions can
be compared
to yeah any last comments or um Robert
with last
word yeah well many thanks to all of you
for for your comments
um it's been most useful and I
guess I just finally request if I I
would like to have to get some
commentary from all sorts of people on
this hypothesis so if you would send me
some comments just initial ones and gut
reaction comments if you like uh on this
core question of can we dismiss the wave
hypothesis or do we need to take it
seriously then that would uh kind of
kick off some of the commentary anyway
many thanks
thank you to all the guests and thank
you Robert all right farewell thank you
bye everyone bye thank you